

ChE 321 – Chemical Reaction Engineering – Fall 2025

Student Learning Outcome 1

Rubric for assessment problem on the midterm computational exam (Problem #1)

Application of mass balances to a plug-flow reactor with recycle to improve conversion.

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem MATLAB Implementation 25 pts	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (<10 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (10 - 15 pts)	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (> 15 pts)
4. Translates academic theory into engineering applications Problem Analysis 40 pts	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (< 10 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (10 - 22 pts)	Does not appear to grasp the connection between theory and the problem (> 22 pts)
6. The uses of appropriate resources needed to solve problems Problem Using ODE Solvers 20 pts	Uses no resources to solve problems (<5 pts)	Uses limited resources to solve problems (5 - 10 pts)	Uses appropriate resources to locate information needed to solve problems (> 10 pts)

Averages:

Performance Indicators: (2) 2.55 (4) 2.45 (6) 2.82 Overall Average: 2.61

The average scores for PI 2 and PI 4 are similar, but the scores for PI 6 is slightly higher and improved from last cycle of review. This is because the instructor has developed examples of fully converged simulations in MATLAB and provided tutorials on selecting solvers and convergence criteria to get numerically correct solutions. For each concept taught in the class, there was a follow-up live-session to code and simulate reaction engineering problems. The live examples were very helpful not only in visualizing the theoretical concepts taught in the class but also in problem-solving for home assignments and exams.

The score for PI 4 was slightly lower because some students were unable to fully converge the solution and thus missed the opportunity to analyze the data. In the future, PI will create problems so that relevant PIs can be evaluated without them being dependent on each other.

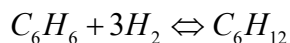
CHE 321 – Written & Computer Exam

(Closed Notes, Closed Books)

Time: 9:30 am to 12:00 pm

Name: _____

Question: The following elementary gas-phase reaction is to be carried out isothermally in a tubular packed-bed reactor with pressure drop. The inlet pressure is 8 atm.



a) Combine the mole balance, rate law, and stoichiometry to develop a set of ordinary differential equations for conversion X and pressure p .

*[Rubric: 15 points for writing mole balance, 10 points for implementing rate law and stoichiometry, and 5 points for deriving governing ODEs; **Total points: 30**]*

b) Solve the ordinary differential equations in MATLAB for the following set of parameters- $C_{T0} = 8 \text{ mol/dm}^3$, stoichiometric feed where $F_{A0} = 10 \text{ mol/s}$, $k_A = 10 \text{ dm}^9/(\text{mol}^3 \cdot \text{kg cat. s})$, $K_C = 20 \text{ dm}^9/\text{mol}^3$, $\alpha = 0.07 \text{ kg}^{-1}$, and $Dp = 1 \text{ mm}$. Plot concentrations of species and pressure in the reactors for total catalyst weight of 10 kg. Now, plot the exit conversion versus catalyst particle diameter (Dp) ranging from 0.1 mm to 10 mm.

*[Rubric: 30 points for correctly implementing parameters of the problem in MATLAB, 20 points for developing MATLAB codes with necessary debugging, and 20 points for obtaining the plots of conversion vs particle diameter; **Total points: 70**]*

Solution:

```
clc; % clear command window  
clear; close all % clear workspace
```

```
y0=[0 1]; % initial conditions
```

```
wspan=[0,100]; %weight span
```

```

%[w,y]=ode45(@PBR_fun,wspan,y0);
%w is a column vector
%y is a matrix, where columns are the species and rows are the weight
%y=[X(w1) p(w1)]
% [X(w2) p(w2)]
% [X(w3) p(w3)]
% ---- original two plots at reference alpha ----
alpha_ref = 0.02;
[w, y] = ode45(@(w,y) PBR_fun(w, y, alpha_ref), wspan, y0);

figure(1);
plot(w,y(:,1));
xlabel('Weight');
ylabel('Conversion');

figure(2)
plot(w,y(:,2));
xlabel('Weight');
ylabel('p');

% ---- third figure: exit conversion vs catalyst particle diameter ----
Dp = linspace(10, 0.1, 100); % mm
X_exit = zeros(size(Dp));

for i = 1:length(Dp)
    % assumption: alpha inversely proportional to Dp
    alpha_i = alpha_ref * (1 / (Dp(i)) );

    [w_i, y_i] = ode45(@(w,y) PBR_fun(w, y, alpha_i), wspan, y0);

    % exit conversion = conversion at final bed weight
    X_exit(i) = y_i(end, 1);
end

figure(3);
plot(Dp, X_exit, 'LineWidth', 2);
xlabel('Catalyst particle diameter, D_p (mm)');
ylabel('Exit conversion');
title('Exit conversion vs catalyst particle diameter');
grid on;

function h=PBR_fun(w,y, alpha) % t and y are row vectors
X=y(1);
p=y(2);

% setting parameters
cA0=0.2;

```

```

FA0=2;
k=6;
%alpha=0.02;
eps=-2/3;

if p <= 1e-8
    p = 1e-8;
end

% Apply stoichiometry
cA=cA0*(1-X)*p/(1+eps*X);
cB=cA0*(2-2*X)*p/(1+eps*X);

%applying rate law
rA=-k*(cA*cB^2);

%apply relative rates

%applying mole balance
h1=-rA/FA0;
h2=-alpha*(1+eps*X)/(2*p);

h=[h1,h2]';%needs to be column vector

```

Performance Indicators-Specific Grading

PI 2 – Understanding the physical problem – Governing equation and boundary conditions and correctly implementing them on MATLAB to solve for concentration profiles. (25 points)

Unsatisfactory <10 points, Developing <15 points, Satisfactory >15 points

PI 6 – Identify appropriate subroutines to solve differential equations and implement criteria to realize convergence. (20 points)

Unsatisfactory <5 points, Developing <10 points, Satisfactory > 10 points

PI 4 – Analyze the effect of parameters and trends in conversion profiles. (40 points)

Unsatisfactory <10 points, Developing <22 points, Satisfactory > 22 points